Cell G6: Frobenius and 2-torsion structure at m = 9..20

Design note, revision 2, prepared and not run (2026-10-27). Every number below is arithmetic or a unit-test result; no experiment has produced data.

1. Question and instrument

Does any Frobenius or 2-torsion structure change the relation yield of an ordered-slot factor base at toy n, against the same instrument without the structure, at matched slot-size profile? (Revision 2 dropped the cost half, since NI and UP have identical operation counts by construction, and the attribution to m: at fixed toy n the slot size moves with m, so cells with $n = 13, 17$ at $m = 9, 10$ vary it at fixed m.)

The instrument is an exact census. For ordered slots $S_1, ..., S_m$ (slot S_i = points whose x lies in a coordinate subspace V_i, including the 2-torsion point T for $x = 0$) it computes $H[R] = |\{(P_1, ..., P_m) \in S_1 \times ... \times S_m : \sum P_i = R\}|$ for every group element R at once, by two meet-in-the-middle halves and all pairs. It runs no solver, so it cannot show an improvement over the index-calculus baseline and reports no distance to rho.

2. Arms and controls

NI	random normal basis, slot i = conjugate indices $\equiv i \bmod m$ (the `n83_rotated_m10` rule): the Frobenius-structured design
UP	generic basis, each slot redrawn until it has exactly the point count of the paired NI slot: matched profile, no Frobenius structure
UP2 / DUP	independent second UP draw (A/A); UP with two slots made identical (positive control, expected dispersion about 2)
NC / NR / ONB	contiguous and random partitions in a normal basis (unpaired, descriptive); the type-II optimal normal basis at $n = 23$ (one instance)

Decision rule (frozen, implemented in `analyze.rs`): paired log-ratios of yield efficiency, dispersion and the orbit Fano factor; EQUIVALENT by TOST at $\pm 10\%$, DIFFERENT only beyond a Bonferroni critical value and the margin, otherwise UNRESOLVED with the minimum detectable effect. A cell whose positive control, A/A, conservation, planted-tuple or digit-model check fails is VOID. NI/UP pairs whose NI instance is a Frobenius chain are a derived stratum (next section), not a test.

3. What the design work found before any data

1. On-curve status is Frobenius invariant (x is on the curve iff x^2 is), so full interleaved slots are Frobenius rotations of each other. Their sizes are tied together and the factor-base size takes only a few values, each a whole number of bits apart (figure below, hand arithmetic of PROTOCOL section 4.3). At $n = 23$ the production partition rule cannot reach the window $\log_2 \mu \in [-10, 5]$ for $m \geq 16$; those cells are expected UNMATCHABLE, and the census measures that exactly.
2. The translation $P \mapsto P + T$ acts on x as $x \mapsto \frac{\sqrt{b}}{x}$ (proved, tested). For prime n only a one-dimensional subspace is stable, so the 2^{m-1} stabiliser needs non-subspace slots. For k stable slots, $H_\iota[R] = 2^{k-1}\big(H_{\text{plain}}[R] + H_{\text{plain}}[R + T]\big)$: the stable base is the plain base queried at R and $R + T$, at 2^k times the enumeration. Relabelling unless an ι-aware enumerator exists.
3. **(Review, revision 2.)** In the class-(0,1) cells the NI slots are $\{T, \pm\sigma^i P\}$, a chain of consecutive Frobenius images, and $\tau^2 \mp \tau + 2 = 0$ forces collisions among the 3^L digit sums: 9711 distinct of 177147 ($n = 23$, $m = 12$), 2335 of 19683 ($m = 14$), 543 of 2187 ($n = 19$, $m = 12$), matched exactly to the group law and the real builder. NI yield efficiency is then lower by $\times 4$ to $\times 18$ and dispersion higher by $\times 6$ to $\times 31$ as an algebraic fact. The cells report this as CHAIN_DERIVED, cross-checked at run time; it is a toy-slot-size effect and says nothing about production slot size or $n = 131$. The earlier prior "EQUIVALENT within 10 percent" is withdrawn.
4. Orbit-union (symmetric) bases are oversized at toy n by 2^8 ($m = 9$) to 2^{26} ($m = 20$) for a single orbit, and sane only near $n = 131$ (three orbits at $m = 20$). They cannot be enumerated.
5. The chained SAT encoding has $(m - 1)(3n^2 + 4n) + n(2m - 1)$ variables and 2^{m-2} pattern instances (65536 at $m = 18$, not 16384); the source stops at $m = 6$.

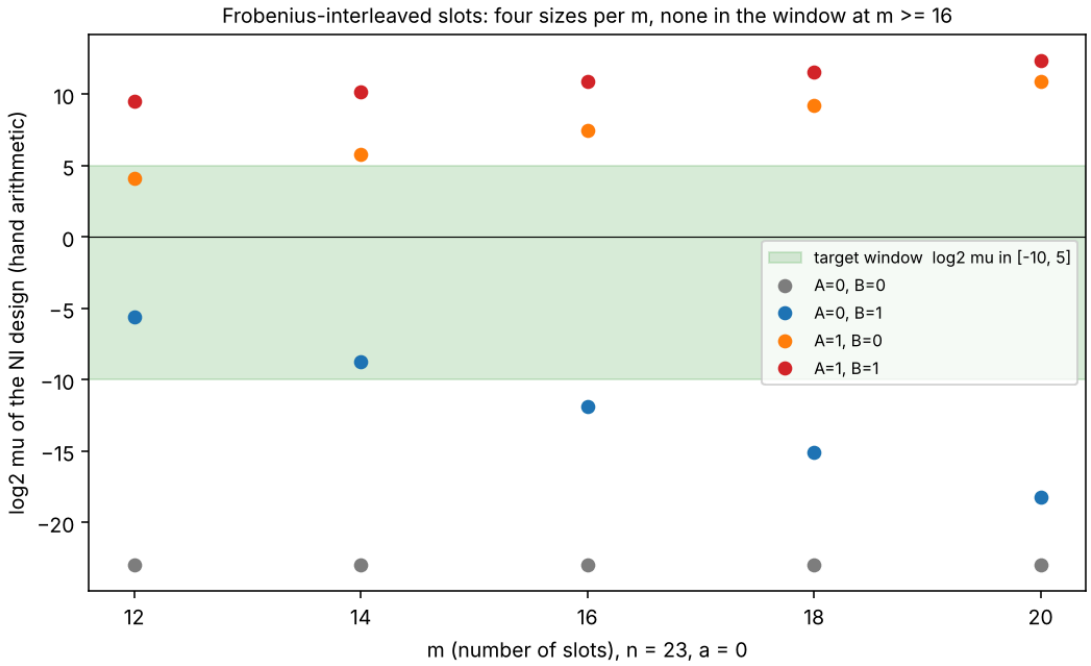


Figure 1: Sizes the Frobenius-interleaved design can take at $n = 23$, $a = 0$ (hand arithmetic: A and B are the on-curve bits of β and $\beta + \beta^{2^m}$).

4. Cannot reach

Solver costs (SAT, Groebner, degree of regularity); $n = 131$; orbit-union bases at $m \geq 9$; Frobenius-stable slots; the variant that excludes $x = 0$; the prime-subgroup stratum; $m \in \{13, 15, 17, 19\}$; the $n = 83$ production gate.

5. Status

Crate `g6-large-m` builds and passes 32 unit tests on tiny fixtures; 26 cells are planned. The S3 resume-checkpoint gate blocks runs in the preparation container; `RUN_CARD.md` lists the exact commands.